In [1]:
```
import pandas as pd
```

In [2]:
```
pd.__version__
```

Out[2]:
```
'2.3.3'
```

In [4]:
```
df= pd.read_excel(r'C:\Users\HP\OneDrive\Documents\Full Stack Data Science With
```

In [5]:
```
df
```

Out[5]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income
1	Country_2	COU1	19.888	12.8	Lower middle income
2	Country_3	COU2	39.245	60.3	High income
3	Country_4	COU3	11.846	65.2	Lower middle income
4	Country_5	COU4	28.864	68.3	Low income
...	...	...	...	...	...
495	Country_496	COU5	16.513	71.5	High income
496	Country_497	COU6	45.140	11.1	Upper middle income
497	Country_498	COU7	41.209	14.2	Lower middle income
498	Country_499	COU8	23.677	89.6	Upper middle income
499	Country_500	COU9	36.754	52.5	High income

500 rows × 5 columns

In [6]:
```
len(df)  # alwasy give rows
```

Out[6]:
```
500
```

In [7]:
```
df.shape # dimension -- rows & columns
```

Out[7]:
```
(500, 5)
```

In [9]:
```
len(df.columns)
```

Out[9]:
```
5
```

In [10]:
```
df.columns  # it will display the column names
```

Out[10]:
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [13]:
```
df
```

Out[13]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income
1	Country_2	COU1	19.888	12.8	Lower middle income
2	Country_3	COU2	39.245	60.3	High income
3	Country_4	COU3	11.846	65.2	Lower middle income
4	Country_5	COU4	28.864	68.3	Low income
...	...	...	...	...	...
495	Country_496	COU5	16.513	71.5	High income
496	Country_497	COU6	45.140	11.1	Upper middle income
497	Country_498	COU7	41.209	14.2	Lower middle income
498	Country_499	COU8	23.677	89.6	Upper middle income
499	Country_500	COU9	36.754	52.5	High income

500 rows × 5 columns

In [14]:
```
df.isnull()
```

Out[14]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
495	False	False	False	False	False
496	False	False	False	False	False
497	False	False	False	False	False
498	False	False	False	False	False
499	False	False	False	False	False

500 rows × 5 columns

In [17]:
```
df.isna()
```

Out[17]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
495	False	False	False	False	False
496	False	False	False	False	False
497	False	False	False	False	False
498	False	False	False	False	False
499	False	False	False	False	False

500 rows × 5 columns

In [19]:
```
df.isnull().sum()
```

Out[19]:
```
CountryName      0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
dtype: int64
```

In [20]:
```
type(df)
```

Out[20]:
```
pandas.core.frame.DataFrame
```

In [21]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   CountryName    500 non-null    object
 1   CountryCode    500 non-null    object
 2   BirthRate      500 non-null    float64
 3   InternetUsers  500 non-null    float64
 4   IncomeGroup    500 non-null    object
dtypes: float64(2), object(3)
memory usage: 19.7+ KB
```

In [24]:
```
df.dtypes
```

Out[24]:
```
CountryName       object
CountryCode       object
BirthRate        float64
InternetUsers    float64
IncomeGroup       object
dtype: object
```

In [25]:
```
df.head()  # system bydefault prints top 5 rows
```

Out[25]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income
1	Country_2	COU1	19.888	12.8	Lower middle income
2	Country_3	COU2	39.245	60.3	High income
3	Country_4	COU3	11.846	65.2	Lower middle income
4	Country_5	COU4	28.864	68.3	Low income

In [26]:
```
df.tail() # system bydefault prints last 5 rows
```

Out[26]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
495	Country_496	COU5	16.513	71.5	High income
496	Country_497	COU6	45.140	11.1	Upper middle income
497	Country_498	COU7	41.209	14.2	Lower middle income
498	Country_499	COU8	23.677	89.6	Upper middle income
499	Country_500	COU9	36.754	52.5	High income

In [27]:
```
df.head(2)
```

Out[27]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income
1	Country_2	COU1	19.888	12.8	Lower middle income

indexing and Slicing in Pandas

In [28]:
```
df
```

Out[28]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income
1	Country_2	COU1	19.888	12.8	Lower middle income
2	Country_3	COU2	39.245	60.3	High income
3	Country_4	COU3	11.846	65.2	Lower middle income
4	Country_5	COU4	28.864	68.3	Low income
...	...	...	...	...	...
495	Country_496	COU5	16.513	71.5	High income
496	Country_497	COU6	45.140	11.1	Upper middle income
497	Country_498	COU7	41.209	14.2	Lower middle income
498	Country_499	COU8	23.677	89.6	Upper middle income
499	Country_500	COU9	36.754	52.5	High income

500 rows × 5 columns

In [29]:
```
df[:]
```

Out[29]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income
1	Country_2	COU1	19.888	12.8	Lower middle income
2	Country_3	COU2	39.245	60.3	High income
3	Country_4	COU3	11.846	65.2	Lower middle income
4	Country_5	COU4	28.864	68.3	Low income
...	...	...	...	...	...
495	Country_496	COU5	16.513	71.5	High income
496	Country_497	COU6	45.140	11.1	Upper middle income
497	Country_498	COU7	41.209	14.2	Lower middle income
498	Country_499	COU8	23.677	89.6	Upper middle income
499	Country_500	COU9	36.754	52.5	High income

500 rows × 5 columns

In [31]:
```
df[5:9]
```

Out[31]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
5	Country_6	COU5	23.999	45.4	Lower middle income
6	Country_7	COU6	38.733	42.4	Lower middle income
7	Country_8	COU7	15.674	24.8	Upper middle income
8	Country_9	COU8	14.881	88.4	Lower middle income

In [34]:
```
df[3:90:2]
```

Out[34]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
3	Country_4	COU3	11.846	65.2	Lower middle income
5	Country_6	COU5	23.999	45.4	Lower middle income
7	Country_8	COU7	15.674	24.8	Upper middle income
9	Country_10	COU9	25.671	87.4	Upper middle income
11	Country_12	COU1	10.235	82.2	High income
13	Country_14	COU3	22.307	83.1	Low income
15	Country_16	COU5	16.488	16.6	Low income
17	Country_18	COU7	27.249	38.3	High income
19	Country_20	COU9	17.050	21.0	Upper middle income
21	Country_22	COU1	46.100	68.4	Upper middle income
23	Country_24	COU3	43.550	27.3	Lower middle income
25	Country_26	COU5	38.079	9.2	Upper middle income
27	Country_28	COU7	23.917	85.6	Low income
29	Country_30	COU9	10.561	77.3	Low income
31	Country_32	COU1	40.381	42.6	Upper middle income
33	Country_34	COU3	25.124	34.0	Lower middle income
35	Country_36	COU5	46.064	14.2	Lower middle income
37	Country_38	COU7	31.757	66.9	Lower middle income
39	Country_40	COU9	45.306	51.9	High income
41	Country_42	COU1	28.028	60.2	High income
43	Country_44	COU3	14.728	11.4	High income
45	Country_46	COU5	24.250	45.3	Upper middle income
47	Country_48	COU7	39.593	58.2	Upper middle income
49	Country_50	COU9	13.539	1.5	Lower middle income
51	Country_52	COU1	21.811	83.0	Low income
53	Country_54	COU3	12.245	81.2	Low income
55	Country_56	COU5	28.008	72.6	Lower middle income
57	Country_58	COU7	20.156	49.4	Lower middle income
59	Country_60	COU9	11.341	87.5	Lower middle income
61	Country_62	COU1	28.828	16.6	Upper middle income
63	Country_64	COU3	23.100	41.0	Upper middle income
65	Country_66	COU5	15.985	86.9	High income
67	Country_68	COU7	28.509	65.9	Lower middle income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
69	Country_70	COU9	13.486	2.5	Low income
71	Country_72	COU1	47.581	27.8	High income
73	Country_74	COU3	39.962	45.6	Upper middle income
75	Country_76	COU5	47.516	1.4	Upper middle income
77	Country_78	COU7	41.272	85.5	Lower middle income
79	Country_80	COU9	27.817	32.1	High income
81	Country_82	COU1	25.754	83.4	High income
83	Country_84	COU3	20.808	91.5	Upper middle income
85	Country_86	COU5	17.376	2.0	High income
87	Country_88	COU7	10.825	47.6	High income
89	Country_90	COU9	46.390	58.1	High income

In [38]:
```
df[-456:]
```

Out[38]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
44	Country_45	COU4	49.265	74.0	High income
45	Country_46	COU5	24.250	45.3	Upper middle income
46	Country_47	COU6	13.786	92.5	Low income
47	Country_48	COU7	39.593	58.2	Upper middle income
48	Country_49	COU8	40.476	74.4	Upper middle income
...	...	...	...	...	...
495	Country_496	COU5	16.513	71.5	High income
496	Country_497	COU6	45.140	11.1	Upper middle income
497	Country_498	COU7	41.209	14.2	Lower middle income
498	Country_499	COU8	23.677	89.6	Upper middle income
499	Country_500	COU9	36.754	52.5	High income

456 rows × 5 columns

In [39]:
```
df['CountryName']
```

Out[39]:
```
0        Country_1
1        Country_2
2        Country_3
3        Country_4
4        Country_5
          ...
495    Country_496
496    Country_497
497    Country_498
498    Country_499
499    Country_500
Name: CountryName, Length: 500, dtype: object
```

In [43]:
```
df[['BirthRate','InternetUsers']]
```

Out[43]:

	BirthRate	InternetUsers
0	17.049	77.2
1	19.888	12.8
2	39.245	60.3
3	11.846	65.2
4	28.864	68.3
...	...	...
495	16.513	71.5
496	45.140	11.1
497	41.209	14.2
498	23.677	89.6
499	36.754	52.5

500 rows × 2 columns

Split the Data into Caterogerica & Numerical

In [44]:
```
df.head(1)
```

Out[44]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income

In [45]:
```
df_cat = df[['CountryName', 'CountryCode','IncomeGroup']]
df_cat.head(1)
```

Out[45]:

	CountryName	CountryCode	IncomeGroup
0	Country_1	COU0	High income

In [47]:
```
print(df.shape)
print(df_cat.shape)
```

```
(500, 5)
(500, 3)
```

In [48]:
```
print(df.columns)
print()
print(df_cat.columns)
print()
```

```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')

Index(['CountryName', 'CountryCode', 'IncomeGroup'], dtype='object')
```

In [49]:
```
df.describe() # bydefatul describe provide inform atnumerical data only
```

Out[49]:

	BirthRate	InternetUsers
count	500.000000	500.000000
mean	29.547558	47.797400
std	12.046262	26.979339
min	8.007000	1.400000
25%	18.639750	24.300000
50%	29.756500	47.650000
75%	40.041500	71.425000
max	49.879000	95.000000

In [50]:
```
df['InternetUsers'] < 2
```

Out[50]:
```
0      False
1      False
2      False
3      False
4      False
       ...
495    False
496    False
497    False
498    False
499    False
Name: InternetUsers, Length: 500, dtype: bool
```

In [51]:
```
df[df['BirthRate'] > 40]
```

Out[51]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
14	Country_15	COU4	41.868	3.4	Lower middle income
21	Country_22	COU1	46.100	68.4	Upper middle income
23	Country_24	COU3	43.550	27.3	Lower middle income
31	Country_32	COU1	40.381	42.6	Upper middle income
35	Country_36	COU5	46.064	14.2	Lower middle income
...	...	...	...	...	...
479	Country_480	COU9	40.052	73.2	Low income
484	Country_485	COU4	40.752	60.2	Upper middle income
491	Country_492	COU1	46.516	76.4	Upper middle income
496	Country_497	COU6	45.140	11.1	Upper middle income
497	Country_498	COU7	41.209	14.2	Lower middle income

126 rows × 5 columns

In [52]:
```
df[(df.BirthRate > 40) & (df.InternetUsers < 2)]
```

Out[52]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
75	Country_76	COU5	47.516	1.4	Upper middle income

In [53]:
```
df['IncomeGroup'] == 'High income'
```

Out[53]:
```
0       True
1      False
2       True
3      False
4      False
       ...
495     True
496    False
497    False
498    False
499     True
Name: IncomeGroup, Length: 500, dtype: bool
```

In [54]:
```
df['IncomeGroup'].unique()
```

Out[54]:
```
array(['High income', 'Lower middle income', 'Low income',
       'Upper middle income'], dtype=object)
```

In [55]:
```
df['IncomeGroup'].nunique()
```

Out[55]:
```
4
```

In [57]:
```
df.columns = ['a','b','c','d','e']
df.columns
```

Out[57]:
```
Index(['a', 'b', 'c', 'd', 'e'], dtype='object')
```

In [58]:
```
df.columns = ['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers','Income
df.head(2)
```

Out[58]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income
1	Country_2	COU1	19.888	12.8	Lower middle income

In [59]:
```
df.BirthRate * df.InternetUsers
```

Out[59]:
```
0      1316.1828
1       254.5664
2      2366.4735
3       772.3592
4      1971.4112
          ...
495    1180.6795
496     501.0540
497     585.1678
498    2121.4592
499    1929.5850
Length: 500, dtype: float64
```

In [60]:
```
df['myCalc'] = df.BirthRate * df.InternetUsers
```

In [63]:
```
df['myCalc']
```

Out[63]:
```
0      1316.1828
1       254.5664
2      2366.4735
3       772.3592
4      1971.4112
          ...
495    1180.6795
496     501.0540
497     585.1678
498    2121.4592
499    1929.5850
Name: myCalc, Length: 500, dtype: float64
```

In [64]:
```
df = df.drop('myCalc',axis = 1)
```

In [65]:
```
df
```

Out[65]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Country_1	COU0	17.049	77.2	High income
1	Country_2	COU1	19.888	12.8	Lower middle income
2	Country_3	COU2	39.245	60.3	High income
3	Country_4	COU3	11.846	65.2	Lower middle income
4	Country_5	COU4	28.864	68.3	Low income
...	...	...	...	...	...
495	Country_496	COU5	16.513	71.5	High income
496	Country_497	COU6	45.140	11.1	Upper middle income
497	Country_498	COU7	41.209	14.2	Lower middle income
498	Country_499	COU8	23.677	89.6	Upper middle income
499	Country_500	COU9	36.754	52.5	High income

500 rows × 5 columns

In [ ]: